Sergey Krivosheev

Data Engineer

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Experience

Data Engineer Jul 2024 – Present

Nexign — Leading Russian OSS/BSS software vendor for Billing and Telecom

- Maintained and developed **ETL pipelines** in **Apache Airflow**, integrating external **APIs**, **HDFS**, **Spark**, **SFTP**, and **S3**. Improved validation task performance by **3×**, reducing DAG execution windows and lowering compute costs for scheduled workloads
- Migrated legacy **Spark** jobs from **Java RDD**-based implementations to **Scala** using the **Spark DataFrame API**. Reduced cluster resource consumption by **30%** and execution time by **80%**, increasing cluster throughput and avoiding additional infrastructure scaling. Engineered new functionality using **Spark UDFs**
- Researched and modified **Spark** source code to extend platform capabilities, including integrating **Spark Streaming** with **SFTP** and enhancing the **Spark JDBC** connector for nullable enum inserts into **PostgreSQL**
- Optimized **ClickHouse** queries and refactored table storage schemas in ETL jobs, reducing execution time by **30–40%** and decreasing analytical workload pressure on production systems
- Researched and integrated **Flink** for implementing **stateful streaming jobs**. Developed jobs in Java using the **DataStream API** with custom sink operators supporting **exactly-once** semantics. Configured deployment for a self-managed job cluster using **Ansible**
- Deployed and administered an **Arenadata Dist Hadoop** cluster with **Hive** and **Spark** in a staging environment. Migrated **HDFS** data and the **Hive Metastore** from a legacy cluster, preventing downtime and mitigating risks during platform upgrades
- Successfully completed major versions migrations: **Python 3.6 → 3.12**, **Airflow 2 → 3**, **Spark 2 → 3**, **Java 8 → 21**, **Spring Boot 2 → 3**, **Hadoop 2 → 3**, reducing technical debt and long-term support costs
- Developed several **Kotlin** services as MVPs. Increased **RPS** can be processed by implementing **multi-instance, thread-safe concurrency** with **at-least-once** semantics, enabling horizontal scaling by X times and vertical scaling by 25%. Enhanced transparency by improving service observability, including detailed logs, tracing, and comprehensive monitoring
- Maintained and extended **Java/Kotlin** backend services by fixing production issues and implementing new features. Refactored legacy codebases to improve developer experience, reducing time-to-market for new features by **25%** and lowering ongoing maintenance effort

Skills

Programming Languages: Python, Kotlin, Java, Scala

Data & Data Engineering: SQL, PostgreSQL, ClickHouse, Airflow, Pandas, NumPy, Hadoop Ecosystem (HDFS, YARN, Hive), Spark (Batch & Streaming), Flink, Kafka, S3/Ceph

JVM & Java Frameworks: Java 8/21, Spring Boot (MVC, Security, Data JPA/Hibernate, AOP), Concurrency, JUnit, Mockito, Gradle

Python Frameworks & Tools: FastAPI, Celery, SQLAlchemy, Psycopg, Pytest, Poetry, Flake8, Black, Mypy

Backend & DevOps Tools: Nginx, Redis, Prometheus, Grafana, Telegraf, Docker, RPM, Bash, Linux, WireMock, Ansible, GitLab CI

Languages: English (C1), Russian (Native)

Education

Higher School of Economics (HSE University), Faculty of Computer Science Sep 2023 – Jun 2025

Master's degree in Machine Learning and High Load Systems *GPA: 9.87/10*

Thesis: Solar Magnetic Tornado Detection Using Neural Networks

Higher School of Economics (HSE University) Sep 2019 – Jun 2023

Bachelor's degree in Economics *GPA: 8.78/10*